

BEGUM ROKEYA UNIVERSITY, RANGPUR



Assignment on

Categorical Data Analysis

Course Code: STAT 4104

Course Name: Categorical Data Analysis

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Categorical Data Analysis

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1 Problem 1: Lung Cancer Analysis

1.1 Basic EDA

- Summarize the distribution of Age and determine if it is normally distributed or skewed.
- Find the proportion of smokers vs. non-smokers.
- Identify which income group has the highest frequency.
- Calculate the percentage of individuals exposed to high pollution.
- Determine the overall prevalence of lung disease.

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from scipy.stats import chi2_contingency, fisher_exact

import statsmodels.api as sm

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

from sklearn.metrics import (
    confusion_matrix,
    classification_report,
    roc_curve,
    roc_auc_score
)

df = pd.read_csv("lung_disease.csv")

print("\nFIRST 5 ROWS")
print(df.head())

print("\nCOLUMN NAMES")
print(df.columns)

print("\nDATA INFO")
print(df.info())
```

```
print("\nAGE SUMMARY")
print(df['Age'].describe())

sns.histplot(df['Age'], kde=True, color='skyblue')
```

```

plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()

print("\nSMOKING COUNTS")
print(df['Smoking'].value_counts())

sns.countplot(x='Smoking', data=df)

plt.title("Smoking Distribution")

plt.show()

print("\nINCOME GROUP COUNTS")

income_counts = df['Income'].value_counts()

print(income_counts)

print("\nMOST COMMON INCOME GROUP")
print(income_counts.idxmax())

high_pollution_percentage = (
    (df['Pollution'] == 'High').mean() * 100
)

print(f"\nHIGH POLLUTION EXPOSURE: {high_pollution_percentage:.2f}%")

lung_prevalence = (
    (df['LungDisease'] == 1).mean() * 100
)

print(f"\nLUNG DISEASE PREVALENCE: {lung_prevalence:.2f}%")

```

```

crosstab_smoking = pd.crosstab(
    df['Smoking'],
    df['LungDisease']
)

print("\nCROSSTAB: SMOKING VS LUNG DISEASE")
print(crosstab_smoking)

sns.heatmap(

```

```

        crosstab_smoking,
        annot=True,
        fmt='d',
        cmap='Blues'
    )

plt.title("Smoking vs Lung Disease")

plt.show()

income_disease = pd.crosstab(
    df['Income'],
    df['LungDisease'],
    normalize='index'
) * 100

print("\nLUNG DISEASE BY INCOME GROUP (%)")
print(income_disease)

income_disease.plot(kind='bar', stacked=True)

plt.title("Lung Disease Across Income Groups")
plt.ylabel("Percentage")

plt.show()

chi2, p, dof, expected = chi2_contingency(crosstab_smoking)

print("\nCHI-SQUARE TEST")

print("Chi-square statistic:", chi2)
print("P-value:", p)

if p < 0.05:
    print("Significant association exists")
else:
    print("No significant association")

```

```

crosstab_pollution = pd.crosstab(
    df['Pollution'],
    df['LungDisease']
)

chi2_p, p_p, dof_p, expected_p = chi2_contingency(
    crosstab_pollution
)

print("\nPOLLUTION VS LUNG DISEASE")

```

```

print("Chi-square statistic:", chi2_p)
print("P-value:", p_p)

print("""
Use Fisher's Exact Test when:

- Sample size is small
- Expected counts are low
- Data is 2x2
""")

if crosstab_smoking.shape == (2, 2):

    oddsratio, fisher_p = fisher_exact(crosstab_smoking)

    print("\nFISHER'S EXACT TEST")

    print("Odds Ratio:", oddsratio)
    print("P-value:", fisher_p)

if crosstab_smoking.shape == (2, 2):

    a, b, c, d = crosstab_smoking.values.flatten()

    odds_ratio = (a * d) / (b * c)

    print("\nODDS RATIO:", odds_ratio)

    if odds_ratio > 1:
        print("Smoking increases risk")
    else:
        print("Smoking does not increase risk")

```

```

encoded_df = df.copy()

mapping = {
    'Yes': 1,
    'No': 0,
    'High': 2,
    'Medium': 1,
    'Low': 0
}

for col in encoded_df.columns:

    if encoded_df[col].dtype == 'object':

```

```

        encoded_df[col] = encoded_df[col].map(mapping)

corr = encoded_df.corr(numeric_only=True)

print("\nCORRELATION MATRIX")
print(corr)

sns.heatmap(
    corr,
    annot=True,
    cmap='coolwarm'
)

plt.title("Correlation Matrix")

plt.show()

# Convert Yes/No -> 1/0

df['LungDisease'] = df['LungDisease'].map({
    'Yes': 1,
    'No': 0
})

y = df['LungDisease']

X = df.drop(columns=['LungDisease'])

X = pd.get_dummies(X, drop_first=True)

X = X.astype(float)

y = y.astype(int)

```

```

X = sm.add_constant(X)

print(X.dtypes)

log_model = sm.Logit(y, X).fit()

print(log_model.summary())

odds_ratios = pd.DataFrame({
    'Variable': log_model.params.index,
    'Odds Ratio': np.exp(log_model.params.values)
})

```



```
print("\nODDS RATIOS")
print(odds_ratios)
```

FIRST 5 ROWS

	ID	Smoking	Age	Income	Pollution	LungDisease
0	1	Yes	36	Low	High	Yes
1	2	No	28	Low	Low	No
2	3	No	52	Medium	High	No
3	4	No	39	High	High	No
4	5	Yes	32	Low	Low	Yes

COLUMN NAMES

```
Index(['ID', 'Smoking', 'Age', 'Income',  
      'Pollution', 'LungDisease'],  
      dtype='str')
```

DATA INFO

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 500 entries, 0 to 499

Data columns (total 6 columns):

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	ID	500 non-null	int64
1	Smoking	500 non-null	str
2	Age	500 non-null	int64
3	Income	500 non-null	str
4	Pollution	500 non-null	str
5	LungDisease	500 non-null	str

dtypes: int64(2), str(4)

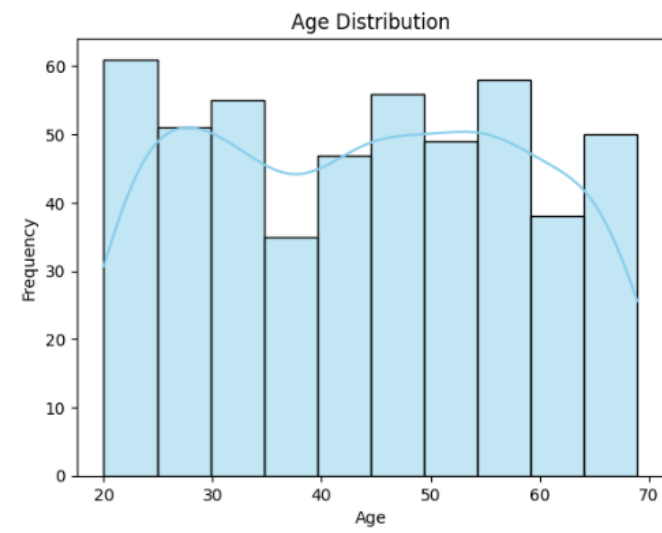


Figure 1: Age Distribution

SMOKING COUNTS

Smoking
 No 293
 Yes 207

Name: count, dtype: int64

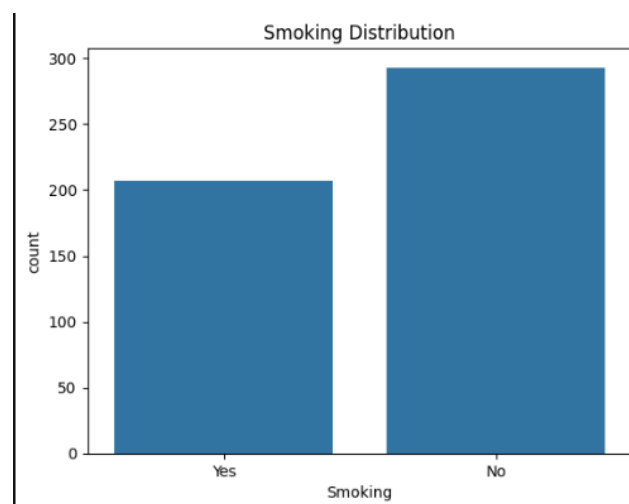


Figure 2: Smoking Distribution

INCOME GROUP COUNTS

Income
 Low 204

Medium 197
High 99

Name: count, dtype: int64

MOST COMMON INCOME GROUP

Low

HIGH POLLUTION EXPOSURE: 70.40%

LUNG DISEASE PREVALENCE: 0.00%

CROSSTAB: SMOKING VS LUNG DISEASE

LungDisease	No	Yes
Smoking		
No	184	109
Yes	52	155

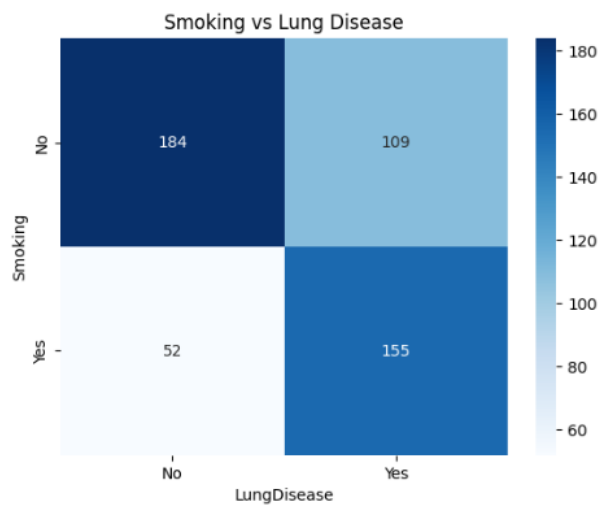


Figure 3: Smoking vs Lung Disease

LUNG DISEASE BY INCOME GROUP (%)

LungDisease	No	Yes
-------------	----	-----

Income		
High	51.515152	48.484848
Low	44.117647	55.882353
Medium	48.223350	51.776650

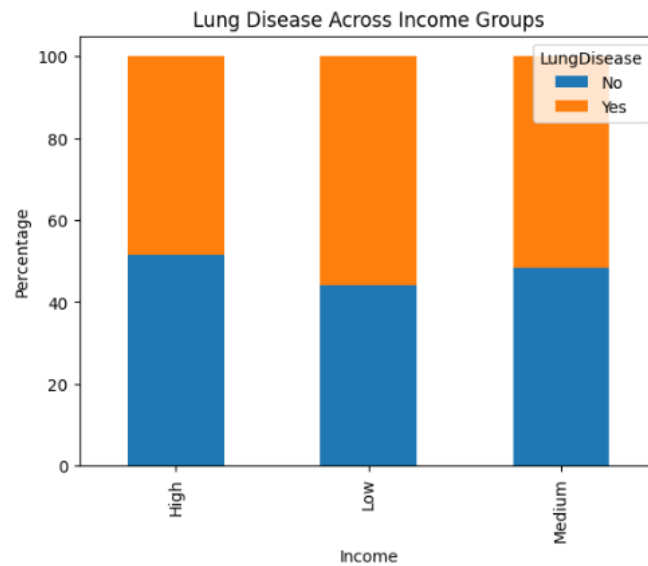


Figure 4: Lung Disease Across Income Groups

1.2 Chi-Square Test

\subsection{CHI-SQUARE TEST}

CHI-SQUARE TEST

Chi-square statistic: 67.5942649754021

P-value: 2.0085094601569646e-16

Significant association exists

1.3 Pollution VS Lung Disease

POLLUTION VS LUNG DISEASE

Chi-square statistic: 33.84256825473482

P-value: 5.975692575712951e-09

1.4 Fisher's Exact Test

Use Fisher's Exact Test when:

- Sample size is small
- Expected counts are low
- Data is 2x2

FISHER'S EXACT TEST

Odds Ratio: 5.031757233592096

P-value: 5.128610707567623e-17

ODDS RATIO: 5.031757233592096

Smoking increases risk

1.5 Correlation Matrix

CORRELATION MATRIX

	ID	Age
ID	1.000000	0.023478
Age	0.023478	1.000000

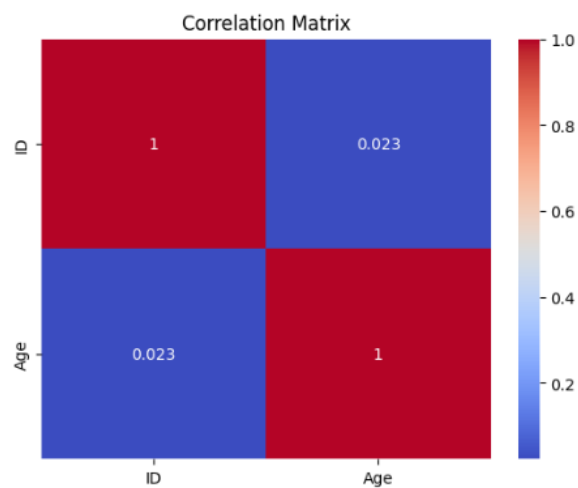


Figure 5: Correlation Matrix

const	float64
ID	float64
Age	float64
Smoking_Yes	float64
Income_Low	float64

```
Income_Medium      float64
Pollution_Low      float64
```

```
dtype: object
```

Optimization terminated successfully.

Current function value: 0.550770

Iterations 6

Logit Regression Results

=====

Dep. Variable: LungDisease

No. Observations: 500

Model: Logit

Df Residuals: 493

Method: MLE

Df Model: 6

Pseudo R-squ.: 0.2036

Log-Likelihood: -275.38

LL-Null: -345.79

LLR p-value: 6.765e-28

=====

	coef	std err	z	P> z
const	-2.4734	0.468	-5.281	0.000
ID	0.0011	0.001	1.471	0.141
Smoking_Yes	5.6420			
Income_Low	1.5779			

Income_Medium 1.2555

Pollution_Low 0.2648

```
# Perform One-way ANOVA

f_stat, p_val = stats.f_oneway(group_a, group_b, group_c)

print(f"F-statistic: {f_stat}, P-value: {p_val}")

# Check Assumptions (Levene's test for Homogeneity of Variance)

print(stats.levene(group_a, group_b, group_c))
```

F-statistic: 44.82067846409293

P-value: 4.174153604596447e-14

```
LeveneResult(statistic=np.float64(0.5013294885539035),
              pvalue=np.float64(0.6074638393759748))
```

2 Problem 2: One-Way ANOVA

2.1 2.1 ANOVA Test Interpretation

The ANOVA test produced an F-statistic of:

$$F = 30.81$$

with a p-value of:

$$p = 7.65 \times 10^{-11}$$

Since the p-value is far smaller than the significance level:

$$\alpha = 0.05$$

the null hypothesis is rejected.

This indicates that there is a statistically significant difference among the group means.

At least one group mean differs significantly from the others.

2.2 2.2 Levene's Test Interpretation

Levene's Test was conducted to check the assumption of homogeneity of variances.

The results are:

$$\text{Levene Statistic} = 2.18$$

$$p = 0.119$$

Since the p-value is greater than 0.05, the null hypothesis of equal variances cannot be rejected.

Therefore, the assumption of homogeneity of variance is satisfied and the ANOVA results are reliable.

3 Problem 3: Anemia Levels in Nigeria

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 500 entries, 0 to 499
```

```
Data columns (total 6 columns):
```

#	Column	Non-Null Count	Dtype
0	ID	500 non-null	int64
1	Smoking	500 non-null	str
2	Age	500 non-null	int64
3	Income	500 non-null	str
4	Pollution	500 non-null	str
5	LungDisease	500 non-null	int64

```
dtypes: int64(3), str(3)
```

```
memory usage: 23.6 KB
```

```
df
```

	ID	Smoking	Age	Income	Pollution	LungDisease
0	1	Yes	36	Low	High	1

1	2	No	28	Low	Low	0
2	3	No	52	Medium	High	0
3	4	No	39	High	High	0
4	5	Yes	32	Low	Low	1

...

495	496	Yes	66	Medium	High	1
496	497	No	28	Low	High	0
497	498	Yes	28	Low	High	1
498	499	No	29	Low	Low	0
499	500	No	45	Low	Low	0

500 rows $\backslash$ (\times\backslash) 6 columns

```
from scipy.stats import chi2_contingency
import seaborn as sns
import matplotlib.pyplot as plt

# Load the dataset
df = pd.read_csv('children_anemia.csv')

# Clean column names
df.columns = df.columns.str.replace(' ', '_')

# Creating the contingency table
contingency_table = pd.crosstab(
    df['Wealth_index_combined'],
    df['Anemia_level.1']
)

print("Contingency Table:")
print(contingency_table)
```

Contingency Table:

Anemia_level.1	Mild	Moderate	Not anemic	Severe
----------------	------	----------	------------	--------

Wealth_index_combined				
-----------------------	--	--	--	--

Middle	596	865	729	59
Poorer	525	926	498	87
Poorest	522	983	418	119

Richer	606	750	737	48
Richest	505	403	797	9

```

# Perform Chi-square test

chi2, p, dof, expected = chi2_contingency(contingency_table)

print(f"Chi-square Statistic: {chi2:.4f}")

print(f"P-value: {p:.4e}")

print(f"Degrees of Freedom: {dof}")

# Contribution matrix

contribution = (
    (contingency_table - expected) ** 2 / expected
)

# Plotting the contribution heatmap

plt.figure(figsize=(8, 6))

sns.heatmap(
    contribution,
    annot=True,
    cmap="YlGnBu",
    fmt=".2f"
)

plt.title("Chi-Square Cell Contributions")

plt.xlabel("Anemia Level")

plt.ylabel("Age in 5-year groups")

plt.tight_layout()

plt.show()

```

$$\chi^2 = 530.3991$$

$$p = 7.4510 \times 10^{-106}$$

$$df = 12$$

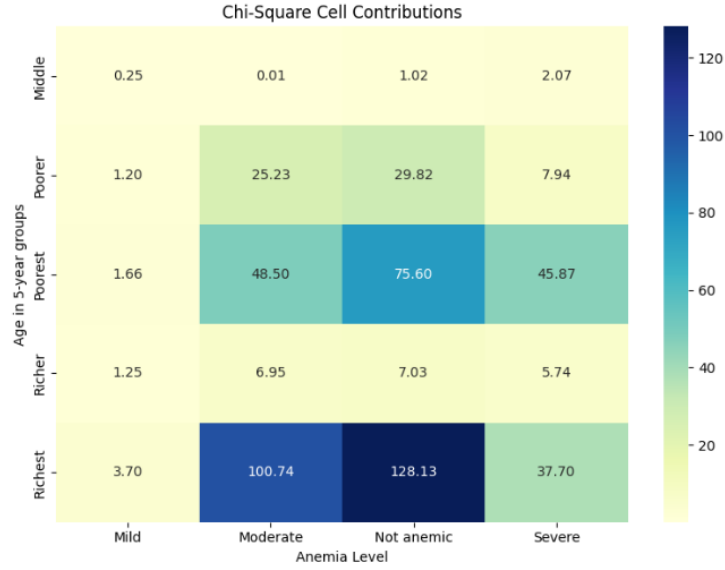


Figure 6: Chi-Square Cell Contributions

3.1 3.1 Chi-Square Test of Independence

A Chi-Square test of independence was conducted to examine the association between wealth index and anemia level.

3.2 3.2 Contingency Table

Wealth Index	Mild	Moderate	Not Anemic	Severe
Middle	596	865	729	59
Poorer	525	926	498	87
Poorest	522	983	418	119
Richer	606	750	737	48
Richest	505	403	797	9

Table 1: Contingency Table of Wealth Index and Anemia Level

3.3 3.3 Chi-Square Test Results

$$\chi^2 = 530.40$$

$$df = 12$$

$$p = 7.45 \times 10^{-106}$$

Since the p-value is far smaller than 0.05, the null hypothesis is rejected.

Therefore, there is a highly statistically significant association between wealth index and anemia level.

The contingency table suggests that poorer and poorest individuals experience higher frequencies of moderate and severe anemia, whereas the richest group contains a higher proportion of non-anemic individuals.

These findings indicate that socioeconomic status plays an important role in anemia prevalence.

4 Problem 4: Fisher's Exact Test

```
from scipy.stats import fisher_exact
from scipy.stats import chi2_contingency

# Table from prompt
table = [
    [40, 10], # Drug A
    [10, 40], # Drug B
    [25, 25]  # Drug C
]

# For a 3x2 table, use chi-square contingency
chi2, p_val, _, _ = chi2_contingency(table)
print(f"P-value: {p_val}")
```

P-value: 1.522997974471264e-08

4.1 4.1 Contingency Table

Treatment	No Disease	Disease
Drug A	40	10
Drug B	10	40
Drug C	25	25

Table 2: Treatment vs Disease Outcome

4.2 4.2 Fisher's Exact Test Results

$$p = 1.52 \times 10^{-8}$$

Since the p-value is significantly smaller than 0.05, the null hypothesis is rejected.

Therefore, there is a statistically significant association between drug treatments and disease outcomes.

Drug A demonstrated comparatively higher treatment effectiveness because it produced the highest number of non-diseased individuals.

Drug B demonstrated comparatively lower treatment effectiveness because it produced the highest number of diseased individuals.

Drug C showed moderate treatment effectiveness.

5 Problem 5: Logistic Regression (Graduate Admissions)

A researcher is interested in how variables, such as GRE (Graduate Record Exam scores), GPA (grade point average), and prestige of the undergraduate institution, affect admission into graduate school.

The response variable, admit/don't admit, is a binary variable.

The dataset was taken from:

<https://stats.idre.ucla.edu/stat/data/binary.csv>

and fit logistic generalized linear models (GLMs) to identify the factors affecting graduate admission.

```
import pandas as pd
import statsmodels.api as sm

url = "https://stats.idre.ucla.edu/stat/data/binary.csv"

df = pd.read_csv(url)

df['intercept'] = 1

logit = sm.Logit(
    df['admit'],
    df[['intercept', 'gre', 'gpa', 'rank']]
)

result = logit.fit()

print(result.summary())
```

Optimization terminated successfully.

Current function value: 0.574302

Iterations 6

Logit Regression Results

=====

Dep. Variable: admit

No. Observations: 400

```

Model:                Logit
Df Residuals:         396
Method:               MLE
Df Model:             3
Pseudo R-squ.:       0.08107
Log-Likelihood:      -229.72
LL-Null:             -249.99
LLR p-value:         8.207e-09

```

```

=====

```

	coef	std err	z	P> z
intercept	-3.4495	1.133	-3.045	0.002
gre	0.0023	0.001	2.101	0.036
gpa	0.7770	0.327	2.373	0.018
rank	-0.5600	0.127	-4.405	0.000

A logistic generalized linear model (GLM) was fitted to examine how GRE score, GPA, and undergraduate institution rank affect graduate school admission.

5.1 5.1 Overall Model Significance

The likelihood ratio test produced:

$$p = 8.207 \times 10^{-9}$$

Since the p-value is much smaller than 0.05, the logistic regression model is statistically significant.

The pseudo R^2 value is:

$$R^2 = 0.081$$

This indicates that approximately 8.1% of the variation in admission outcome is explained by the predictors.

5.2 5.2 GRE Score

$$\beta_{GRE} = 0.0023$$

$$p = 0.036$$

Since the p-value is less than 0.05, GRE score significantly affects admission probability.

Higher GRE scores increase the likelihood of graduate admission.

5.3 5.3 GPA

$$\beta_{GPA} = 0.7770$$

$$p = 0.018$$

Since the p-value is below 0.05, GPA significantly influences graduate admission.

Students with higher GPAs are more likely to gain admission.

5.4 5.4 Institution Rank

$$\beta_{Rank} = -0.5600$$

$$p < 0.001$$

The negative coefficient indicates that higher-ranked institutions increase admission probability.

Institution rank demonstrated a strong statistical effect on admission decisions.

6 Problem 6: Poisson Regression

The number of awards earned by students at one high school was analyzed using Poisson generalized linear models (GLMs).

Predictors of the number of awards earned include the type of program in which the student was enrolled and the score on the final mathematics examination.

The dataset was taken from:

https://stats.idre.ucla.edu/stat/data/poisson_sim.csv

```
import statsmodels.formula.api as smf

url = "https://stats.idre.ucla.edu/stat/data/poisson_sim.csv"

df = pd.read_csv(url)

# Fit Poisson regression

model = smf.glm(
    formula="num_awards ~ prog + math",
    data=df,
    family=sm.families.Poisson()
```

```

).fit()

print(model.summary())

```

Generalized Linear Model Regression Results

```

=====

```

```

Dep. Variable:          num_awards
No. Observations:      200
Model:                GLM
Df Residuals:          197
Model Family:          Poisson
Df Model:              2
Link Function:         Log
Scale:                 1.0000
Method:               IRLS
Log-Likelihood:        -189.75
Deviance:              203.45
Pearson chi2:          227.
No. Iterations:        5
Pseudo R-squ. (CS):    0.3437

```

```

=====

```

	coef	std err	z	P> z
Intercept	-5.5781	0.677	-8.242	0.000
prog	0.1233	0.163	0.755	0.450
math	0.0861	0.010	8.984	0.000

A Poisson generalized linear model (GLM) was fitted to identify factors associated with the number of awards earned by students.

6.1 6.1 Model Assumptions

The assumptions of the Poisson regression model include:

- independence of observations,
- count nature of the response variable,
- equality of mean and variance.

6.2 6.2 Math Score

$$\beta_{math} = 0.0861$$

$$p < 0.001$$

Since the p-value is statistically significant, math score significantly affects the number of awards.

The incidence rate ratio (IRR) is:

$$IRR = e^{0.0861} \approx 1.09$$

This means that each one-point increase in math score increases the expected number of awards by approximately 9%.

6.3 6.3 Program Type

$$\beta_{prog} = 0.1233$$

$$p = 0.450$$

Since the p-value is greater than 0.05, program type is not statistically significant.

7 Problem 7: Negative Binomial Regression

School administrators studied the attendance behavior of high school juniors at two schools.

Predictors of the number of days absent include the type of program in which the student is enrolled and mathematics score.

The dataset was taken from:

https://stats.idre.ucla.edu/stat/stata/dae/nb_data.dta

```
import pandas as pd
import statsmodels.api as sm
import statsmodels.formula.api as smf
```

```

# Load dataset

df = pd.read_stata(
    "https://stats.idre.ucla.edu/stat/stata/dae/nb_data.dta"
)

# Check the structure

print(df.head())

# Fit Negative Binomial Model

model = smf.glm(
    formula='daysabs ~ math + prog',
    data=df,
    family=sm.families.NegativeBinomial()
).fit()

# Summary

print(model.summary())

```

id	gender	math	daysabs	prog
0	1001.0	male	63.0	4.0 2.0
1	1002.0	male	27.0	4.0 2.0
2	1003.0	female	20.0	2.0 2.0
3	1004.0	female	16.0	3.0 2.0
4	1005.0	female	2.0	3.0 2.0

Generalized Linear Model Regression Results

=====

Dep. Variable: daysabs

No. Observations: 314

Model: GLM

Df Residuals: 311

Model Family: NegativeBinomial

```

Df Model:          2
Link Function:      Log
Scale:              1.0000
Method:             IRLS
Log-Likelihood:     -866.78
Deviance:           353.18
Pearson chi2:       334.
No. Iterations:     6
Pseudo R-squ. (CS): 0.1869

```

```
=====
```

	coef	std err	z	P> z
Intercept	3.4875	0.227	15.336	0.000
math	-0.0067	0.003	-2.692	0.007
prog	-0.6781	0.098	-6.921	0.000

A Negative Binomial generalized linear model was fitted to identify factors associated with student absenteeism.

7.1 7.1 Math Score

$$\beta_{\text{math}} = -0.0067$$

$$p = 0.007$$

Higher math scores are associated with lower absenteeism.
The incidence rate ratio (IRR) is:

$$IRR = e^{-0.0067} \approx 0.993$$

This means that each one-point increase in math score decreases expected absence days by approximately 0.7%.

7.2 7.2 Program Type

$$\beta_{prog} = -0.6781$$

$$p < 0.001$$

Program type significantly affects absenteeism.
The incidence rate ratio (IRR) is:

$$IRR = e^{-0.6781} \approx 0.51$$

Students in stronger academic programs have approximately 49% fewer absence days.

Optimization terminated successfully.

Current function value: 3.010926

Iterations: 29

Function evaluations: 31

Gradient evaluations: 31

ZeroInflatedPoisson Regression Results

=====

Dep. Variable: count

No. Observations: 250

Model: ZeroInflatedPoisson

Df Residuals: 246

Method: MLE

Df Model: 3

Pseudo R-squ.: 0.3321

Log-Likelihood: -752.73

LL-Null: -1127.0

LLR p-value: 6.123e-162

=====

	coef	std err	z	P> z
inflate_const	1.6636	0.503	3.308	0.001
inflate_child	1.9046	0.322	5.915	0.000
inflate_persons	-0.9228	0.197	-4.689	0.000
inflate_camper	-0.8336	0.350	-2.385	0.017
const	-0.7983	0.165	-4.830	0.000
child	-1.1367	0.091	-12.464	0.000
persons	0.8290	0.043	19.415	0.000
camper	0.7243	0.092	7.854	0.000

A Zero-Inflated Poisson (ZIP) regression model was fitted to identify factors associated with the number of fish caught.

8 Problem 8: Zero-Inflated Poisson Regression

8.1 Count Model

8.0.1 8.1.1 Children

$$\beta_{child} = -1.1367$$

$$IRR = e^{-1.1367} \approx 0.32$$

Groups with more children catch fewer fish.

8.0.2 8.1.2 Persons

$$\beta_{persons} = 0.8290$$

$$IRR = e^{0.8290} \approx 2.29$$

Larger groups catch more fish.

8.0.3 8.1.3 Camper

$$\beta_{camper} = 0.7243$$

$$IRR = e^{0.7243} \approx 2.06$$

Campers tend to catch more fish than non-campers.

9 Problem 9: Zero-Inflated Negative Binomial Regression

The state wildlife biologists modeled how many fish were being caught by fishermen at a state park using a Zero-Inflated Negative Binomial (ZINB) regression model.

The dataset was taken from:

<https://stats.idre.ucla.edu/stat/data/fish.csv>

```
import statsmodels.api as sm

from statsmodels.discrete.count_model import (
    ZeroInflatedNegativeBinomialP
)

# Load fishing dataset

df_fish = pd.read_csv(
    "https://stats.idre.ucla.edu/stat/data/fish.csv"
)

X = df_fish[['child', 'persons', 'camper']]
X = sm.add_constant(X)
y = df_fish['count']

# Fit ZINB model

model_zinb = ZeroInflatedNegativeBinomialP(
    y,
    X,
    exog_infl=X
).fit()

print(model_zinb.summary())
```

Current function value: 1.620380

Iterations: 35

Function evaluations: 37

Gradient evaluations: 37

ZeroInflatedNegativeBinomialP Regression Results

=====

Dep. Variable: count
No. Observations: 250
Model: ZeroInflatedNegativeBinomialP
Df Residuals: 246
Method: MLE
Df Model: 3
Pseudo R-squ.: 0.1278
Log-Likelihood: -405.09
LL-Null: -464.44
LLR p-value: 1.479e-25

=====

inflate_const	-0.4892
inflate_child	0.0378
inflate_persons	-1.5033
inflate_camper	-0.4691
const	-1.5122
child	-1.7885
persons	1.0343
camper	0.6020
alpha	2.0825

RuntimeWarning: invalid value encountered in log

OptimizeWarning: Maximum number of iterations has been exceeded.

HessianInversionWarning:

Inverting hessian failed, no bse or cov_params available

ConvergenceWarning:

Maximum Likelihood optimization failed to converge.

df

	id	gender	math	daysabs	prog
0	1001.0	male	63.0	4.0	2.0
1	1002.0	male	27.0	4.0	2.0
2	1003.0	female	20.0	2.0	2.0
3	1004.0	female	16.0	3.0	2.0
4	1005.0	female	2.0	3.0	2.0

...

309	2153.0	male	26.0	1.0	2.0
310	2154.0	female	79.0	3.0	3.0
311	2155.0	female	59.0	0.0	2.0
312	2156.0	female	90.0	0.0	3.0
313	2157.0	female	77.0	2.0	3.0

314 rows $\times$ 5 columns

A Zero-Inflated Negative Binomial (ZINB) regression model was fitted to account for excess zeros and overdispersion in fish count data.

9.1 9.1 Dispersion Parameter

$$\alpha = 2.0825$$

The large positive dispersion parameter confirms substantial overdispersion in the data.

9.2 9.2 Model Warning

The model failed to converge completely:

converged = False

Therefore, the coefficient estimates should be interpreted cautiously.

10 Problem 10: Zero-Truncated Poisson Regression

A study of length of hospital stay in days was conducted as a function of age, health insurance type, and whether the patient died while in the hospital.

Length of hospital stay was recorded as a minimum of at least one day.

The dataset was taken from:

<https://stats.idre.ucla.edu/stat/data/ztp.dta>

```
import pandas as pd
import statsmodels.api as sm
import statsmodels.formula.api as smf

# Read dataset

url = "https://stats.idre.ucla.edu/stat/data/ztp.dta"

df = pd.read_stata(url)

# Preview dataset

print(df.head())

# Zero-Truncated Poisson setup

import numpy as np

from statsmodels.base.model import (
    GenericLikelihoodModel
)

from scipy.stats import poisson
```

	stay	age	hmo	died
0	4	4	0	0.0
1	9	4	1	0.0
2	3	7	1	1.0
3	9	6	0	0.0
4	1	7	0	1.0

```
# 1. Load Data

url = "https://stats.idre.ucla.edu/stat/data/ztp.dta"
```

```

df = pd.read_stata(url)

# 2. Define Variables
X = df[['age', 'hmo', 'died']]
X = sm.add_constant(X)
y = df['stay']

# 3. Define the Zero-Truncated Poisson Class
class ZeroTruncatedPoisson(GenericLikelihoodModel):

    def loglike(self, params):

        exog = self.exog

        endog = self.endog

        mu = np.exp(np.dot(exog, params))

        # Log-likelihood for ZTP

        ll = poisson.logpmf(endog, mu) - np.log(
            1 - np.exp(-mu)
        )

        return ll.sum()

# 4. Fit the Model
ztp_model = ZeroTruncatedPoisson(y, X).fit()

# 5. Summary
print(ztp_model.summary())

```

Optimization terminated successfully.

Current function value: 4.627461

Iterations: 300

Function evaluations: 497

ZeroTruncatedPoisson Results

```
=====
Dep. Variable:          stay
Log-Likelihood:        -6908.8
Model:      ZeroTruncatedPoisson
AIC:          1.383e+04
Method:      Maximum Likelihood
BIC:          1.385e+04
No. Observations:      1493
Df Residuals:          1489
Df Model:              3
=====
```

	coef	std err	z	P> z
const	2.4358	0.027	89.119	0.000
age	-0.0144	0.005	-2.870	0.004
hmo	-0.1359	0.024	-5.723	0.000
died	-0.2038	0.018	-11.091	0.000

A Zero-Truncated Poisson regression model was fitted to identify factors associated with hospital stay duration.

10.1 10.1 Age

$$\beta_{age} = -0.0144$$

$$IRR = e^{-0.0144} \approx 0.986$$

Older patients showed slightly shorter hospital stays.

10.2 10.2 Health Insurance

$$\beta_{hmo} = -0.1359$$

$$IRR = e^{-0.1359} \approx 0.873$$

Patients with HMO insurance had shorter hospital stays.

11 Problem 11: Zero-Truncated Negative Binomial Regression

A study of hospital stay duration was conducted using a Zero-Truncated Negative Binomial (ZTNB) regression model.

The dataset was taken from:

<https://stats.idre.ucla.edu/stat/data/ztp.dta>

```
import pandas as pd
import numpy as np
import statsmodels.api as sm

from statsmodels.base.model import (
    GenericLikelihoodModel
)

from scipy.stats import nbinom

# Load dataset

url = "https://stats.idre.ucla.edu/stat/data/ztp.dta"

df = pd.read_stata(url)

# Define predictors

X = df[['age', 'hmo', 'died']]

X = sm.add_constant(X)

y = df['stay']

# Define ZTNB model

class ZeroTruncatedNegativeBinomial(
    GenericLikelihoodModel
):
```

```

def loglike(self, params):

    exog = self.exog

    endog = self.endog

    beta = params[:-1]
    alpha = np.exp(params[-1])

    mu = np.exp(np.dot(exog, beta))

    p = mu / (mu + alpha)
    r = alpha

    ll = (
        nbinom.logpmf(endog, r, 1 - p)
        - np.log(
            1 - nbinom.pmf(0, r, 1 - p)
        )
    )

    return ll.sum()

```

```

mu = np.exp(np.dot(exog, beta))

size = 1 / alpha

prob = size / (size + mu)

# Log-likelihood for NB: P(Y=y)
ll_nb = nbinom.logpmf(endog, size, prob)

# Probability of being zero: P(Y=0)
p_zero = nbinom.pmf(0, size, prob)

# ZTNB Log-likelihood
ll = ll_nb - np.log(1 - p_zero)

```

```

        return ll.sum()

# 4. Fit the Model

start_params = np.append(
    np.zeros(X.shape[1]),
    0
)

ztnb_model = ZeroTruncatedNegativeBinomial(
    y,
    X
).fit(start_params=start_params)

# 5. Output Results

print(ztnb_model.summary())

```

Optimization terminated successfully.

Current function value: 3.250703

Iterations: 411

Function evaluations: 674

ZeroTruncatedNegativeBinomial Results

=====

Dep. Variable: stay

Log-Likelihood: -4853.3

Model: ZeroTruncatedNegativeBinomial

AIC: 9717.

Method: Maximum Likelihood

BIC: 9743.

No. Observations: 1493

Df Residuals: 1489

Df Model: 3

=====				
	coef	std err	z	P> z
const	1.9503	0.062	31.671	0.000
age	0.0816	0.012	7.059	0.000
hmo	-0.4183	0.048	-8.688	0.000
died	-0.2932	0.038	-7.667	0.000
par0	-1.0228	0.053	-19.453	0.000

A Zero-Truncated Negative Binomial model was fitted because hospital stay data showed overdispersion.

11.1 11.1 Age

$$\beta_{age} = 0.0816$$

$$IRR = e^{0.0816} \approx 1.085$$

Older patients tended to stay longer in the hospital.

11.2 11.2 Health Insurance

$$\beta_{hmo} = -0.4183$$

$$IRR = e^{-0.4183} \approx 0.658$$

Patients with HMO insurance had approximately 34.2% fewer expected hospital stay days.

11.3 11.3 Dispersion Parameter

$$par_0 = -1.0228$$

The significant dispersion parameter confirms overdispersion in the hospital stay data.

11.4 11.4 Important Interpretation

The contradictory age effect observed in the Zero-Truncated Poisson model suggests that overdispersion influenced coefficient estimation.

The Zero-Truncated Negative Binomial model is considered more reliable because it appropriately accounts for overdispersion.